

JEE MAIN REVISION MASTER: PROBABILITY

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 44 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Conditional Probability & Bayes' Theorem	38 PYQs	Conditional & Bayes
Moderate	Probability Distribution & Binomial Distribution	6 PYQs	Random Variables

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Conditional Probability & Bayes' Theorem	$P(A B) = \frac{P(A \cap B)}{P(B)}$, $\quad P(A \cap B) = P(A) \cdot P(B)$ (independent), $\quad P(E_i A) = \frac{P(E_i)P(A E_i)}{\sum P(E_j)P(A E_j)}$
Probability Distribution & Binomial Distribution	$\text{Mean } \mu = \sum x_i p_i$, $\quad \text{Var } \sigma^2 = \sum x_i^2 p_i - \mu^2$, $\quad \text{Binomial: } P(X = r) = {}^n C_r p^r q^{n-r}$, $\quad \text{Mean} = np$, $\quad \text{Var} = npq$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Conditional Probability & Bayes' Theorem (Very High Yield — 38 PYQs)

Core Principle: Multiplication rule of probability, test for statistical independence, law of total probability, and inverse probability via Bayes' theorem.

■ **Caution / JEE Pitfall:** Mutually exclusive events ($A \cap B = \emptyset$ implies $P(A \cap B) = 0$) CANNOT be independent if both have non-zero probabilities!

■ **Speed Trick:** For independent events: $P(A \cup B) = 1 - P(A')P(B')$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Recent studies suggest that 12% of the world population is left handed. Depending on parents' hand usage, the chances of having left handed..." [Ans: N/A]
- [2026 Shift] Q#1 (MCQ): "Probability of at least one of the events A and B occur is 0.6 . If A and B occur simultaneously with probability 0.2 , then $P(\bar{A})$..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "The probability that a man and his wife live after 20 years are $(1/4)$ and $(1/3)$ respectively. The probability that neither..." [Ans: D]

3.2 Probability Distribution & Binomial Distribution (Moderate Yield — 6 PYQs)

Core Principle: Discrete probability distributions, expected value $E(X)$, variance, and Binomial Bernoulli trials ($q = 1 - p$).

■ **Caution / JEE Pitfall:** In a Binomial distribution, the variance (npq) is ALWAYS strictly less than the mean (np) because $q < 1$.

■ **Speed Trick:** Variance formula shortcut: $\text{Var}(X) = E(X^2) - [E(X)]^2$.

Representative Source PYQs:

- **[2024 Shift] Q#1 (MCQ):** "A random variable X has the following probability distribution: X 0 1 2 $P(X)$ $\frac{25}{36}$ k $\frac{1}{36}$ If the mean of the random variable X is ..." **[Ans: N/A]**
- **[2024 Shift] Q#2 (MCQ):** "If a random variable X follows the binomial distribution with parameters $n=5, p$ and $P(X=2)=9P(X=3)$, then p is equal to..." **[Ans: N/A]**
- **[2022 Shift] Q#2 (MCQ):** "Find the mean number of heads in three tosses of a fair coin...." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Recent studies suggest that 12% of the world population is left handed. Depending on parents' hand usage, the chances of having left handed children are as follows: A: Both parents are left handed, chances of having left handed children =24% B: Both parents are right handed, chances of having left handed children =9% C: Father left handed and mother right handed, chances of having left handed children =17% D: Father right handed and mother left handed, chances of having left handed children =22% Given $P(A)=P(B)=P(C)=P(D)=1/4$ and L denotes child is left handed. What is the probability that $P(A|L)$?

- (A) (17 / 100)
- (B) (19 / 25)
- (C) (1 / 3)
- (D) (2 / 3)

Correct Answer: None

Solution Analysis: We need to find $P(A|L)$, which means: Probability that both parents are left handed, given that the child is left handed. Using Bayes' theorem, $P(A|L) = \frac{P(A)P(L|A)}{P(L)}$ Here, $P(A)=P(B)=P(C)=P(D)=(1/4)$ Also, $P(L|A)=24\%=(24/100)$ Now find $P(L)$ using the total probability formula:

$P(L) = P(A)P(L|A) + P(B)P(L|B) + P(C)P(L|C) + P(D)P(L|D)$ Substitute the values: $P(L) = (1/4) \times (24/100) + (1/4) \times (9/100) + (1/4) \times (17/100) + (1/4) \times (22/100)$ Since $(1/4)$ is common, $P(L) = (1/4) \times (24+9+17+22/100)$ $P(L)$

Benchmark Problem 2: KCET 2024 (Q#1)

A random variable X has the following probability distribution: X 0 1 2 $P(X)$ $\frac{25}{36}$ k $\frac{1}{36}$ If the mean of the random variable X is $1/3$, then the variance is

- (A) (1 / 18)
- (B) (5 / 18)
- (C) (7 / 18)
- (D) (11 / 18)

Correct Answer: None

Solution Analysis: X 0 1 2 $P(X)$ $(\frac{25}{36})$ k $(\frac{1}{36})$ ■ $\text{Mean} = \sum_{i=1}^n x_i P(x_i) \Rightarrow (1/3) = 0 \times (\frac{25}{36}) + 1 \times k + 2 \times (\frac{1}{36}) \Rightarrow (1/3) = k + (1/18) \therefore k = (1/3) - (1/18) = (6-1/18) = (5/18)$ $\begin{aligned} \text{Now,} \\ \text{variance} &= \sum_{i=0}^n x_i^2 P_i(x) - (\text{mean})^2 \\ &= (0)^2 \times \frac{25}{36} + (1)^2 \times k + (2)^2 \times \frac{1}{36} - \frac{1}{9} \\ &= 0 + k + \frac{1}{9} - \frac{1}{9} = k = \frac{5}{18} \end{aligned}$